Practical No 2

Create Two Networks Using Star Topology and Add a Resource (Printer) Using Cisco Packet Tracer and Show Packet Transmission

AIM

To design and simulate two computer networks using **star topology** in Cisco Packet Tracer, connect a **printer as a shared resource**, and demonstrate **packet transmission between two hosts**.

THEORY

A computer network is a collection of interconnected devices such as computers, printers, switches, and routers that communicate with each other to share data and resources. Network topology defines the physical or logical arrangement of these devices.

Star Topology

In star topology, every device in the network is connected to a central device, usually a switch. All communication between devices passes through this central switch. This structure provides better performance and easy fault isolation because a failure in one cable or device does not affect the rest of the network.

Star topology is widely used in local area networks (LANs) due to its reliability, scalability, and ease of maintenance. Additional devices can be easily added to the network without affecting existing connections.

Switch

A switch is a networking device that connects multiple devices within a network and forwards data packets only to the intended destination device using MAC addresses. This improves network efficiency and reduces unnecessary traffic.

Router

A router is used to connect two or more different networks. In this experiment, the router connects the two star topology networks and enables communication between them by routing data packets based on IP addresses.

Network Printer

A network printer is a printer connected to the network so that multiple users can access it. It is assigned an IP address, allowing computers to send print jobs through the network just like sending data to another device.

IP Addressing

Each device in the network must have a unique IP address to communicate. IP addressing helps in identifying the source and destination of data packets in the network.

Packet Transmission

When data is sent from one host to another, it is divided into small units called packets. These packets contain source and destination addresses and are transmitted through switches and routers.

Cisco Packet Tracer allows visualization of this process using simulation mode, where the movement of packets can be observed step by step, helping in understanding how data flows in a real network.

TYPES OF TOPOLOGY

1. Bus Topology

In bus topology, all devices are connected to a single central cable called the backbone. Data sent by one device travels along the cable and is received by all other devices.

Advantages:

- Low installation cost
- Requires less cabling
- Easy to implement for small networks

Disadvantages:

- Failure of the backbone cable brings down the entire network
- Difficult to troubleshoot
- Network performance decreases as more devices are added

2. Star Topology

In star topology, all devices are connected to a central device such as a switch or hub. All data communication passes through this central device.

Advantages:

- Easy to install, manage, and expand
- Failure of one device does not affect the rest of the network
- High data transfer speed
- Easy fault isolation

Disadvantages:

- Failure of the central device stops the entire network
- Requires more cabling than bus topology

3. Ring Topology

In ring topology, devices are connected in a closed loop, and data travels in one direction from one device to another.

Advantages:

- Equal access to network resources for all devices
- No data collision

Disadvantages:

- Failure of one device or cable disrupts the entire network
- Difficult to reconfigure

4. Mesh Topology

In mesh topology, every device is connected directly to all other devices.

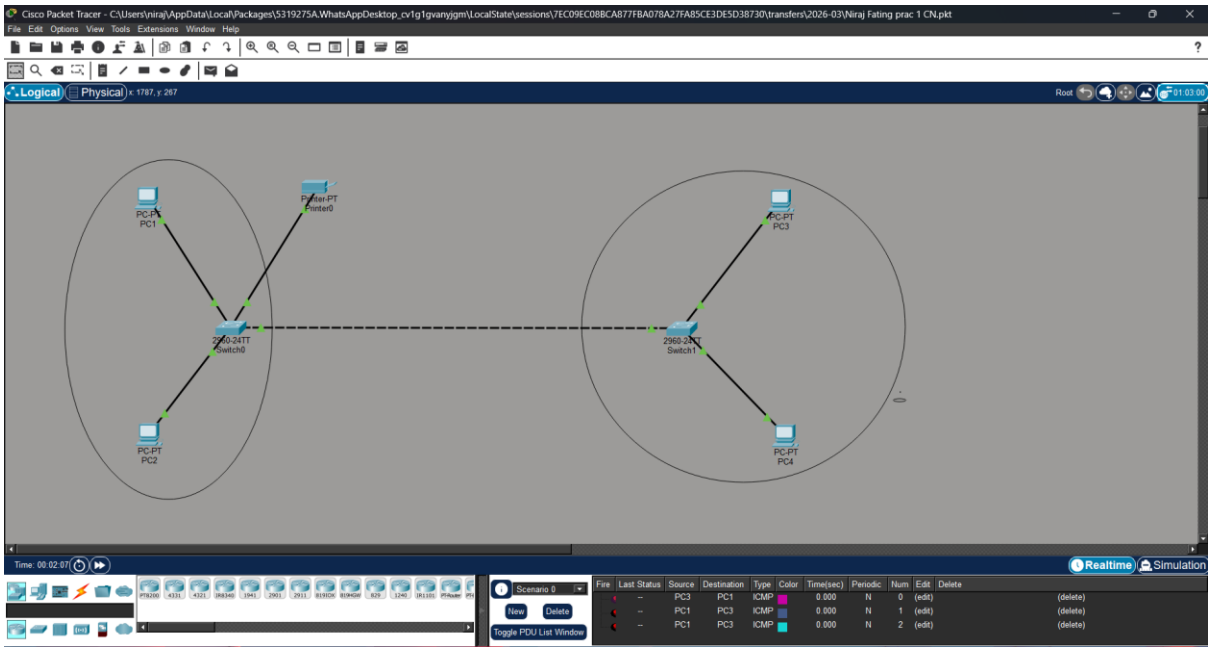
Advantages:

- Highly reliable
- No single point of failure
- Supports heavy data traffic

Disadvantages:

- Very expensive
- Complex cabling and configuration
- Difficult to maintain

DIAGRAM



DETAILS ABOUT DIAGRAM

The given diagram represents **two separate star topology networks** connected together using network switches.

Network 1 (Left Side)

Devices:

- PC1
- PC2
- Printer
- Switch0 (2960-24TT)

Configuration:

PC1, PC2, and the printer are directly connected to Switch0. Switch0 acts as the central connecting device, forming a star topology. The printer is configured as a shared network resource so that multiple computers can access it.

Network 2 (Right Side)

Devices:

- PC3
- PC4
- Switch1 (2960-24TT)

Configuration:

PC3 and PC4 are connected to Switch1, forming another star topology network with Switch1 as the central device.

Interconnection Between Networks

Switch0 and Switch1 are interconnected using a network cable. This connection allows communication between both star networks.

Packet Transmission

When one computer sends data to another:

1. The data is divided into small units called packets.
2. Each packet contains source and destination IP and MAC addresses.
3. The packet is forwarded to the switch.
4. The switch identifies the destination and forwards the packet accordingly.
5. The destination computer receives and reassembles the data.

CONCLUSION

Thus, two networks using star topology were successfully created in Cisco Packet Tracer. A printer was added as a shared network resource, and packet transmission between hosts was observed using simulation mode. This experiment helped in understanding network design, device connectivity, resource sharing, IP addressing, and the flow of data packets in a computer network.

Attendance	Performance	Viva	Total
3	7	5	15